

WASP-43b

R-1525

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-43_210319 · created 2026-08-01T17:34:02+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2459292.8115 +/- 0.002 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.023 +/- 0.056
Transit depth (Rp/Rs)^2	0.053 +/- 0.26 [%]
Orbital Inclination (inc)	77.4 +/- 2.87
Airmass coefficient 1 (a1)	1.42 +/- 0.84
Airmass coefficient 2 (a2)	0.01 +/- 0.79
Scatter in the residuals of the lightcurve fit is	61.81 %
Best Comparison Star	#3 - [307, 279]
Optimal Aperture	1.11
Optimal Annulus	3.5
Transit Duration (day)	0.014 +/- 0.016

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.023 ± 0.056 vs 0.1594 (deviation 2.44σ)
Duration fit vs published	0.014 d vs 0.0512 d (deviation 2.33σ)
Mid-time O–C	-2.5 min
Depth SNR	0.4
Residual scatter	61.81 %

Light curve

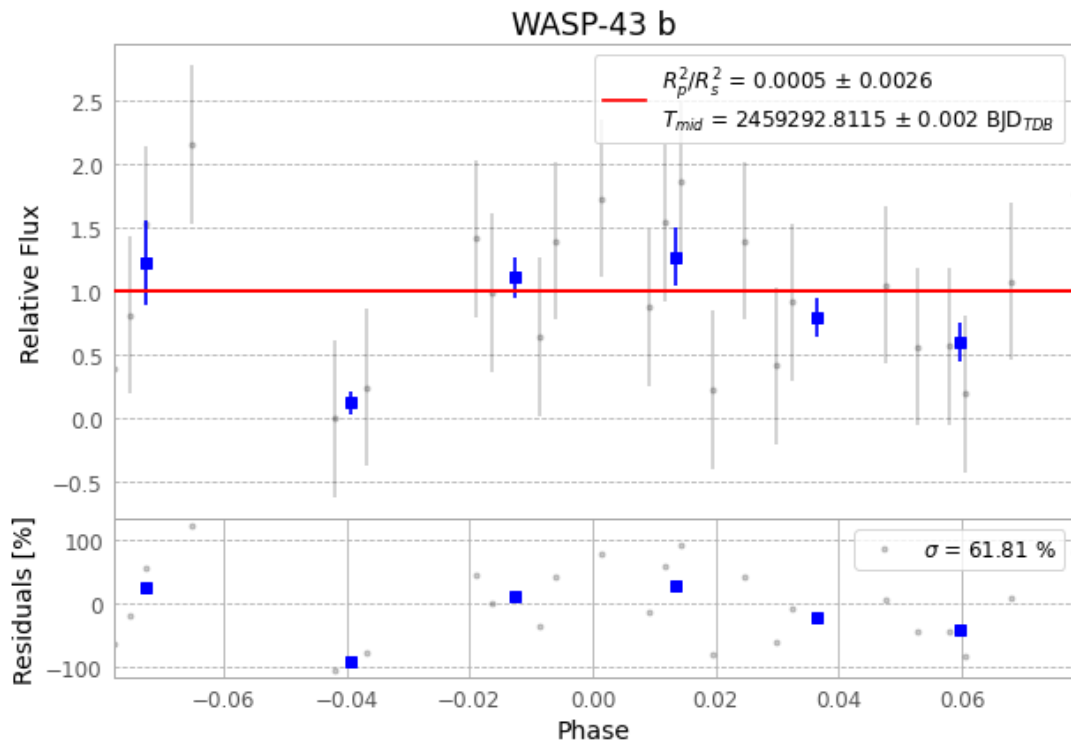


Figure 1 — FinalLightCurve_WASP-43 b_2021-03-18.png (SHA-256 5ac624d9ab1e0ca9...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [301, 336], 3 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 fb96607443ddc958...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

64 FITS frames (42 MB), WASP-43210319054812.FITS → WASP-43210319085712.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-43-210319.log (45fe6f2ddcd3...), FinalLightCurve_WASP-43 b_2021-03-18.png (5ac624d9ab1e...), FinalLightCurve_WASP-43 b_2021-03-18.csv (7b7f7863dc8f...)

Compute resources

Wall time 130.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-01 17:34Z.